

Generative Design: Ideation, Sketching, and Critiquing

In the beginning the Universe was created. This has made a lot of people very angry and been widely regarded as a bad move.

—Douglas Adams, *The Restaurant at the End of the Universe*

Highlights

- Preparing for design via immersion
- The role of synthesis
- Ideas: The building blocks of design
- Ideation:
 - Ideation informers
 - Ideation catalysts
 - Ideation techniques
- Sketching
- Critiquing
- Rules of engagement

13.1 INTRODUCTION

13.1.1 You Are Here

We begin each process chapter with a “You are here” picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 13.1). In this chapter, we get into the UX design process, beginning with generative design.

Compared to the UX research in Part 2, design is a less procedural and more creative activity. While UX research was about observation and understanding things as they currently are, design is about lateral thinking and generating new ideas to make things better.

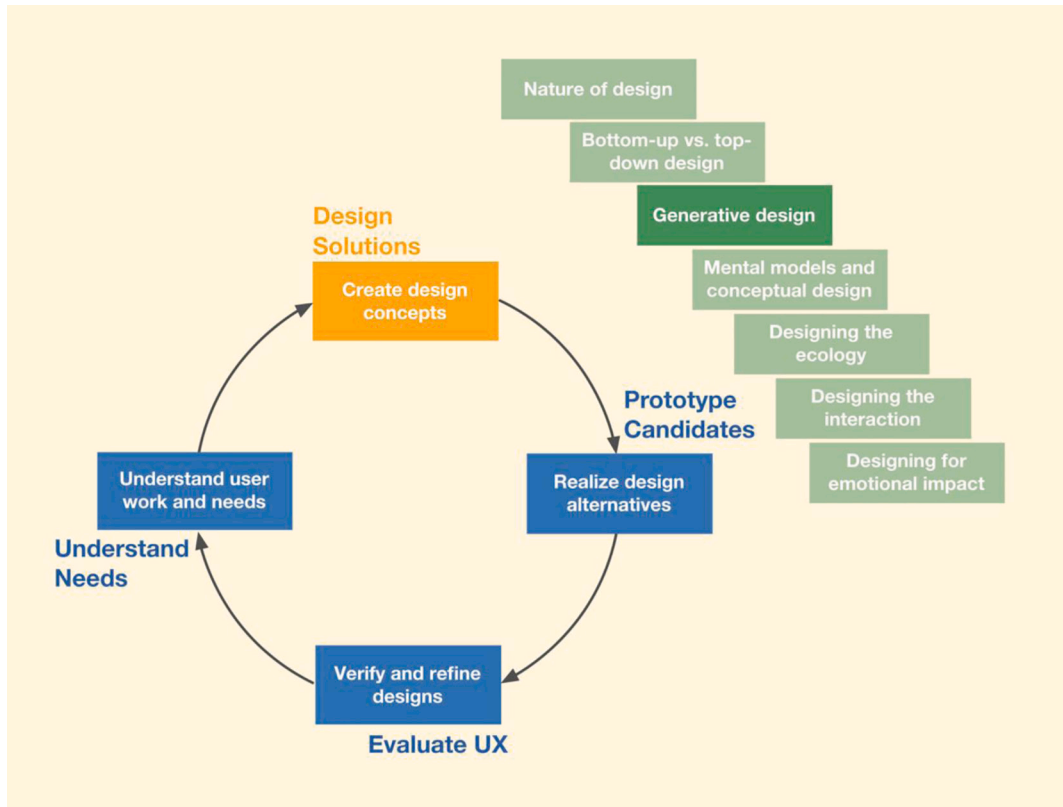


Fig. 13.1

You are here in the generative design chapter for creating UX design concepts within the design solutions activity, in the context of the overall Wheel UX lifecycle.

13.1.2 Preparing for Generative Design: Immersion

Designers, like UX researchers, need immersion—a form of deep thought and analysis—to understand a problem and to make connections among the different aspects of it. To set the stage for ideation, immerse yourself within a design-support milieu, a war room of working artifacts as inputs and inspiration to ideation. Get it all out in front of you to point to, discuss, and critique. Fill your walls, shelves, and worktables with artifacts, representations of ideas, images, visual and tactile props, toys, notes, posters, sketches, diagrams, mood boards (collages of artifacts and images showcasing a theme of emotional impact to be embraced within a UX design [see [Section 17.4.2](#)]), and images of other good designs.

Immersion begins by getting reacquainted with all the models and inputs to design we created during UX research, and it is a pre-requisite to participating in activities for generating new ideas and concepts.

Now the UX studio really shines. The UX studio (see [Section 5.2](#)) is a good place to house your immersive environment. Artifacts trigger ideas and bring to surface the connections and relationships among various entities in the work domain. Designers can refresh their understanding of the work domain by walking up to the appropriate model(s) on the studio walls. The very context of a design studio puts you in a creative frame of mind.

13.1.3 The Role of Synthesis

At its primary level, design is an approach to problem solving based on synthesis, a way of visualizing the full story in a design situation that requires processing multiple inputs and drawing conclusions. From [Chapter 7](#), you may recall the goal of synthesis is to build models and other UX research data representations to explain the problem domain and understand UX design needs. Here, the goal is to build design solutions that involve integrating disparate inputs and what we know about the work domain, users and work practice, together with guidelines, technologies, and constraints to create new designs. The designers must satisfy a variety of technical, business, and cultural constraints and achieve a variety of stakeholder goals. At the same time, designers must manage tradeoffs to create a single unified design to satisfy the ecological, interaction, and emotional needs of users (see [Section 11.2.1](#)).

Synthesis is closely associated with creativity and innovation, to create design ideas that were not present in the UX research data. That is why the early stages of creative design (in this chapter) are called generative design. Design synthesis is “the epitome of the breakthrough idea, the ability to pull together disconnected ideas and arrive at something new and meaningful” (Baty, 2010). In sum, synthesis is a bridge connecting UX research and UX design.

13.1.4 Overview of Generative Design: Intertwining of Ideation, Sketching, and Critiquing

Designers perform generative design in a tightly coupled iterative loop of activities for exploring design ideas ([Fig. 13.2](#)).

These generative design activities include:

- Ideation: the activity where ideas are spawned, a cognitive technique to create varying and innovative design possibilities ([Section 13.3](#))

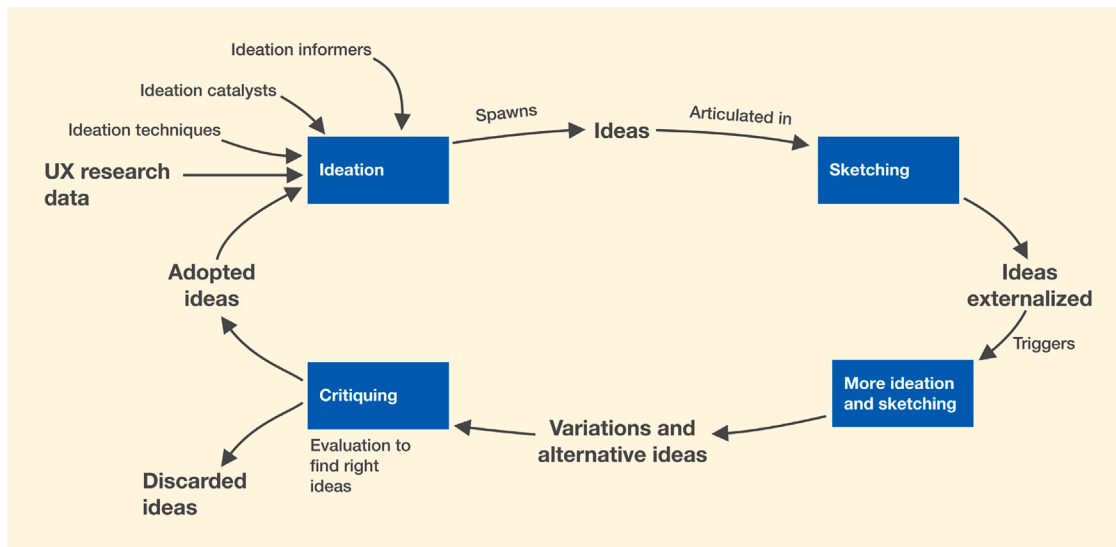


Fig. 13.2

Ideation, sketching, and critiquing activities for generating design ideas.

- Sketching: an externalization activity that captures those ideas in concrete representations ([Section 13.4](#))
- Critiquing: an analysis activity to evaluate the emergent design ideas for tradeoffs ([Section 13.5](#))
- Refining: an activity (usually iterative) where ideas are adopted, modified, or discarded

Even though we describe these activities separately for clarity, they are inextricably intertwined and occur as a rapid iteration of tightly coupled and overlapping activities, often essentially performed simultaneously. Each of these activities supports and stimulates the others. For example, the activity of sketching is a simple and immediate way to capture an idea, which in itself can incite further ideas.

13.2 IDEAS: THE BUILDING BLOCKS OF DESIGN

In this context, an idea, or design idea, is a visualized design proposal that can include visions of new ecologies, interactions, emotional responses, behavior, and capabilities within a system or product.

Ideas are conjured by the human mind or borrowed from observations of nature; they are triggered by focused activity or unexpected catalysts (Section 13.3.4). For example, looking at a breakdown in a flow model can trigger ideas on how to circumvent or mitigate it. Similarly, looking at an inspiring image or artifact in the design studio can trigger a new idea about colors or graphics for the visual design of a product.

Sometimes ideas occur as eureka moments (Section 13.3.4.1), springing up at unexpected places and unexpected times. They can even occur when we are not actively seeking them (e.g., during idea incubation [Section 13.3.5.5]).

Whatever their provenance, ideas by their very nature are delicate and must be cultivated to grow. They should be captured as soon as they occur, or they may be lost forever; they also must be nurtured. Without an environment or culture of openness, ideas can be cut down by premature judgment.

13.3 IDEATION

13.3.1 Creativity in Ideation

If the roof doesn't leak, the architect hasn't been creative enough.

—Frank Lloyd Wright (Donohue, 1989)

Ideation is the most important generative aspect of design; it is the leading edge of the act of creation. Ideation is the process of creating various and innovative proposals for ecological, interaction, and emotional designs. This is a highly creative and fun phase. Diversity in the design team helps with ideation because it brings varied perspectives.

13.3.2 Ideation Informers, Catalysts, and Techniques

There are three types of inputs to ideation: ideation informers, catalysts, and techniques:

- Ideation informers inform design by pointing to information about design requirements and UX research data models to consider in the design (Section 13.3.3).
- Ideation catalysts are design-oriented flashes of inspiration that precipitate creative design ideas (Section 13.3.4).
- Ideation techniques are skills used by designers to foster the spawning and nurturing of design ideas (Section 13.3.5).

13.3.3 Ideation Informers

Ideation informers are informative inputs to consider in the design. They are usually derived from the UX research data and manifest themselves as usage data models and, possibly, the work activity affinity diagram (see [Section 7.4](#)). Informers are not building blocks; you cannot put them together to make a design. Rather, they inform by pointing to design-oriented aspects, such as task descriptions or user personas, to consider in the design.

13.3.3.1 User work roles

User work roles help identify the types of users and work that the design should support and a way to carve out sub-systems in the overall ecology. For example, the ticket buyer work role and the event manager work role help designers think of two separate designs and capabilities. The work these two roles do has almost no overlap, and consequently the designs for these roles will not either. This points to two sub-systems that need to be built, each with different interaction designs and emotional impact aspects but sharing the same overall ecology.

13.3.3.2 Personas

We talked about constructing personas in [Section 8.3.3](#). Personas channel the act of creation toward supporting a particular archetype of a work role in the work domain. Without personas, the inputs designers need to deal with could be overwhelming and often competing or contradictory.

Where personas work best. Personas work best in design situations where the work domain is broad and unconstrained (i.e., projects on the left side of the system complexity space of [Fig. 3.1](#)). When personas are used in designing consumer products or systems in such situations, they help account for the nuances and activities in the personal lives of the various work roles.

In such systems, we can use a synthesis technique in the form of personas to provide inferences that take the conclusion a bit beyond what is explicit in the raw data. For example, Mary is a very busy 35-year-old soccer mom who balances the management of three kids, a household, and a career. You know immediately that a design solution for a calendar management app, for example, cannot require her to focus for long periods in front of a computer. The solution needs to have mobile components, and we need to include various kinds of alerts (calendar alarms, text messages) because she is already overstretched in terms of time management and will be at risk of forgetting important calendar events.

Goals for using personas in design. As we will see, designers often come up with more than one possible user persona for a work role (Cooper, 2004). The idea behind designing for these multiple personas is that the design must make the primary persona very happy while not making any of the selected personas unhappy. Buster (our persona) will love it, and it still works satisfactorily for the others.

Personas provide inputs into emotional aspects because they describe personal traits and preferences of users. Personas also provide a way to evaluate whether the design matches the target user. It is a way to check evolving design plans.

Using personas in design. Start by making your design as though Rachel, your primary persona, is the only user. Team members tell stories about how Rachel would handle a given usage situation. As more and more of her stories are told, Rachel becomes more real and more useful as a medium for conveying requirements.

In Fig. 13.3, let us assume that we have chosen persona P3 as the primary persona out of four selected candidate personas. Because Design(P3) is a design specific to just P3, Design(P3) will work perfectly for P3. Now we have to adjust Design(P3) to make it suffice for P1.

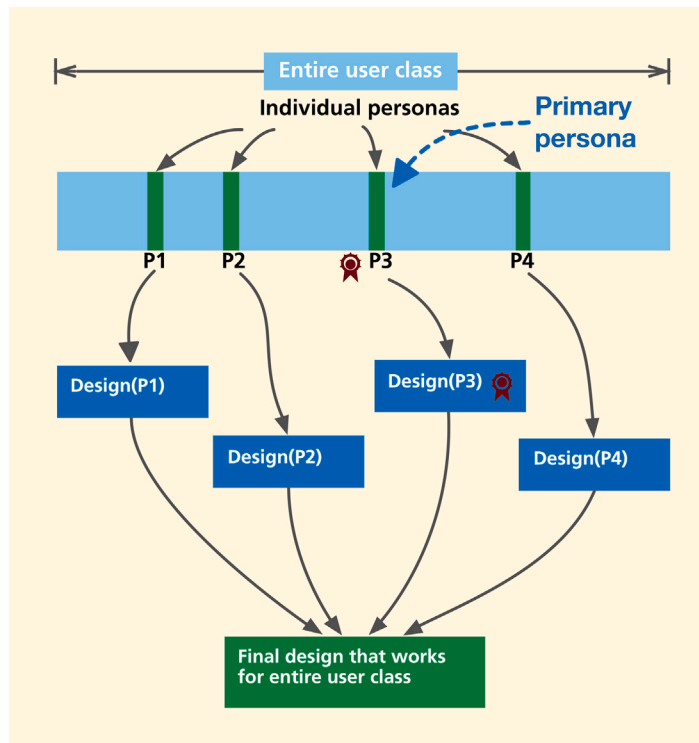


Fig. 13.3

Adjusting a design for the primary persona to work for all the selected personas.

In turn, we adjust it to suffice for P2 and P4. As you converge on the final design, the non-primary personas will be accounted for via Design(P1), Design(P2), and Design(P4), but will defer to the primary persona, P3, in case of conflict. If there is a design tradeoff, you will resolve the tradeoff to benefit the primary persona and still make it work for the other selected personas.

Example: Cooper's In-Flight Entertainment System

Cooper (2004, p. 138) describes the successful use of personas in a Sony design project for an in-flight entertainment system called P@ssport. In addition to the work roles for system maintenance and the flight attendants who set up and operate the entertainment system, the main users are passengers on flights. We call this main work role the Traveler.

The user population that takes on the role of Traveler is about as broad as you can imagine, including essentially all the people who travel by air—almost everyone. Cooper showed how the use of personas helped mitigate the vagueness and diversity of the Traveler user class characteristics.

You could come up with dozens or more personas to represent the Traveler, but in that project the team got it down to four personas, each very different. Three were quite specialized to match the characteristics of a particular type of traveler, and the fourth was more general (i.e., an older guy who was not technology savvy and was not into exploring user interface structures or features—essentially the opposite of most of the characteristics of the other personas).

They considered designs for each of the first three personas, but none of those designs would have worked for the fourth. In the end, they came up with an initial design for the fourth persona and then adapted it to work well for all the other personas without sacrificing its effectiveness for the target persona.

13.3.3.3 Flow models and physical models

Flow models and (in some cases) physical models provide insights into envisioning a new ecology for work practice. Using them as guides, look for ideas to eliminate flows, roles, and redundant data entry. For those flows that cannot be eliminated, look for ideas to make them more efficient and to avoid ecology-level breakdowns and constraints.

13.3.3.4 Task structure and sequence models

In Sections 9.2 and 9.3, we described models for task structure and task sequencing. Look in these models for ideas for interaction design flows. What are different ways to support the different tasks in a sequence model? How

should the design-support interruption of a sequence? What are the best interaction design patterns for each task sequence in these models?

From the task interaction models, look for ideas to reduce and automate steps while avoiding redundant data entry and unnecessary physical motions.

13.3.3.5 Artifact model

Look at the artifacts in the artifact model (see [Section 8.5](#)). Identify metaphors that can relate the system to the artifacts people see and use daily in the work practice.

In [Section 8.5.1](#), we talked about order-form artifacts collected from restaurants. Artifacts can be a great start for an ideation exercise by asking important questions (e.g., How can we use the order-form artifact to make the experience for the customer placing an order in the restaurant more fun, engaging, and informed? or What if each dining table contained an embedded interactive touch screen?). In these cases, diners could browse the menu or read the biosketch of the new chef on the screen, and users could play games or surf the web while waiting to be served.

This interactive table feature can also help overcome an important problem in ordering food: Textual descriptions on a paper menu do not convey the actual eating experience. Paper menus do not leverage the potentially rich human sensual connection to the food offerings! Why not let the customer, using an interactive tabletop or a mobile phone, ask questions about ingredients and see images of the dish being served? Then it is only a small step to let customers place orders themselves.

13.3.3.6 Information architecture model

An information object is an internally stored, structured or simple article or piece of information or data that serves as a work object. Information objects are often data entities central to workflow being operated on by users; they are organized, shared, labeled, navigated, searched and browsed for, accessed and displayed, modified and manipulated, and stored back again, all within a system ecology.

Look for all the information objects in the current work practice that need to be managed in the design. As these information objects move within the envisioned ecology, they are accessed and manipulated by people in various work roles. In an enterprise human resources application, for example, information objects may be employee work history forms and other objects, such as paychecks that are created, modified, and processed by different user roles. Look for ideas on how to structure the various information objects in

the design (i.e., ideas for the information architecture of the objects). Think about what kinds of operations or manipulations will be performed.

What are the different points in the ecology where a certain information object will be accessed? What aspects of that object will be preserved in each of the devices in the ecology? This is where you start defining and laying out the information structure for the ecology. This is called pervasive information architecture (Resmini & Rosati, 2011), which provides ever-present information availability across devices, users, and other parts of a broad ecology.

13.3.3.7 Emotional impact in your design

Look for ideas to help with emotional needs. What are some concerns and issues you identified with the culture and interactor influences in the work practice? Think of ideas to mitigate them in the design. In your UX research data, look for work activity notes about places in the work practice that are drudgery so you can invent fun ways to overcome these feelings. Use these issues as springboards to your design scenarios, sketches, and storyboarding.

13.3.3.8 Social models

The social model, the least commonly used model especially in agile environments, is needed when social and cultural interactions among the people involved in the work practice are complex and/or problematic. The social model captures the culture of the shared workplace, the communal aspects of the users' organizational workplace as a social setting, and the ways they impact how things get done in the work environment. Users are affected by concerns, worries, and job pressures that stem from workplace ambiance and environmental factors, political realities, and power struggles where employees pull rank based on official job titles. A strong indicator of problems in this area are seen in a toxic atmosphere of the workplace, which can have a psychological impact on users. Outside influences that can also stifle productivity include vendors, customers, regulatory agencies, the government, the market, or the competition.

Social model issues crop up when a system is not designed to support a work role or when a work practice built into a UX design runs counter to what users want or need. Or a design may put people in certain work roles in competition with others when cooperation would be more productive.

Even when your UX design is otherwise good, if you ignore the social problems, then you risk the consequences, which can include:

- Built-up resentments by some users
- Lost productivity because of social friction
- Deliberate subversive actions that negatively impact success

Example: Using a Social Model to Resolve a Difference in Goals Between Two Roles

An example of how a social model helped us reason about a misalignment in goals between two roles came from one of our professional experiences (Pardha). In this company, where we built a turnkey private cloud platform, we had two conflicting roles: the sales engineering role and the product development role (including UX and software engineering [SE] roles).

People in the sales engineering role worked with enterprise customers to understand their cloud technology context and use cases. Their focus was on getting a customer to commit to buying the product and were therefore all about upselling current and future capabilities of the product. This often resulted in promises of product capabilities on timelines that were too aggressive for the product development roles. The latter's goals, in addition to everyone's goal of increasing revenue, were about product reliability and quality, and they were directly responsible for metrics related to these concerns.

If an unstable product version was released to customers, then the sales roles would meet their sales goals and get their commissions (a major part of their compensation), but the product development roles (not compensated via commissions) would be left burning the midnight oil to fix high severity bugs. This tension resulted in significant turmoil in the organization. A social model helped us model and reason through this misalignment with the leadership of the company. Subsequent discussion led to structural change in how we built products and to at least one major change in the product. The internal process change was to agree on a guaranteed feature release schedule that was owned by product development roles and for sales to not promise beyond it. To help the sales team with their goals, a new product capability, called tech preview, was launched, allowing customers to try future alpha and beta release candidates of yet unstable capabilities.

Example: Social Model Issues With Middleburg University Ticket Transaction Service

- Ticket buyers:
 - Feel pressure from those in line to complete transaction quickly
 - Worry about the system not completing the transactions correctly or losing tickets or money if the system has an error

- Find the noise and distraction of a public location interferes with thinking and decision making
- Are concerned that they will lose the personal service they like so much if the system is changed to use kiosks
- Ticket sellers:
 - Feels pressure to please customers
 - Can get worn out by the fast pace and hard work of peak periods
 - Worry that complaints about service can ruin job reviews
 - Are subject to influences of the administrative supervisor, who shows up occasionally and causes stress

Find ways, using the social model as a guide, to increase communication, reinforce positive values, address concerns of people in work roles, and accommodate influences.

13.3.3.9 Requirements

In [Chapter 10](#), we talked about capturing design requirements from the UX research data. The statement of a requirement in itself can trigger or inform a design idea. How do other systems support such a requirement? What are some problems or downsides with such an idea? Is there a better way to support that requirement?

13.3.4 Ideation Catalysts

A design catalyst is something that exists or something that happens that leads a designer to spawn a new design idea. An ideation catalyst is not necessarily something a designer can create or willfully plan for and do, but something that arises spontaneously, possibly from brainstorming or storytelling. The designer skill that makes it work is the ability to notice and identify it when it happens. A catalyst can trigger a spark or inspiration that can lead to a eureka moment, unlocking the power of ideation and making it fruitful. Ideation catalysts were important inputs for great inventors, such as Thomas Edison, and great thinkers, such as Albert Einstein.

Example: Design Idea Catalyst: Velcro

A good example of this is seen in the now well-known story (Brown, 2008) of George de Mestral,¹ an engineer who, on a walk in the country, got those little round burrs stuck to his sweater. A non-design thinker in this situation might only get irritated, but de Mestral was thinking like a designer, and he

¹https://en.wikipedia.org/wiki/George_de_Mestral

envisioned a material that could stick together and be pulled apart many times without damage, leading to his invention of Velcro.

The burrs were a design catalyst. They just happened, and that spawned a design idea.

13.3.4.1 The eureka moment

Boling and Smith (2012), citing Krippendorff (2006), described a crucial event in generative design: “When designers work, no matter what process they follow or type of thinking they employ, *they inevitably face the moment of invention* [emphasis ours]. This is the point at which no theory, guideline, example, or statement of best practice can tell the designer or the design team specifically what to do.” This moment of invention is precisely the point where a design idea is created, and something is conceived that did not exist before. A design catalyst facilitates this moment of invention to help spawn the genesis of an idea.

Sometimes ideas occur spontaneously or involve a muse, a kind of catalyst that inspires or guides creativity. It definitely involves insight, instinct, intuition, experience, and a natural ability to think outside the box and see “it,” an aptitude Gladwell (2007) says can be honed in a designer. When “it” happens, something triggers an aha moment, and a design idea emerges. For example, sitting in a bathtub is supposed to have produced the spark that triggered Archimedes’ idea of how to measure the volume of an irregular solid.²

13.3.5 Ideation Techniques

Ideation techniques are skill-based UX practices that UX designers can do to support ideation, sketching, and critiquing while fostering the spawning and nurturing of new design ideas. These techniques are the more general UX design techniques (see [Section 2.3](#)) as applied to ideation. Brainstorming, framing, and storytelling are examples of ideation techniques taken right out of the UX technique toolbox (see [Section 2.3](#)).

13.3.5.1 Framing and reframing

Framing and reframing (see [Section 2.3.3.4](#)) represent a design technique that poses the design problem along a particular dimension to spark a hitherto-unconsidered analysis of tradeoffs. It is a technique to look at an aspect of the work domain along a particular dimension, making it easy to talk about that

²https://en.wikipedia.org/wiki/Archimedes%27_principle

aspect of the problem. Framing acts as scaffolding because it has a structure, and you use that structure to help in analysis.

As Cross (2006) put it: “Designers tend to use solution conjectures as the means of developing their understanding of the problem.” Creating frames is a way to focus not on the generation of solutions but on the ability to create new approaches to the problem situation itself (Dorst, 2015).

Example: An Industrial Engineering Story About Reframing a Problem

This is a story about the production line in a toothpaste factory. Credit goes to an unknown person who started this (possibly apocryphal but illustrative) story circulating across the internet.

The factory had a problem: Production sometimes shipped empty boxes, without the tube inside. This was due to the way the production line was set up, and people with experience in designing production lines will tell you how difficult it is to have everything happen with timing so precise that every single unit coming out of it is perfect. Small variations in the environment (which cannot be controlled in a cost-effective fashion) mean you must have quality assurance checks smartly distributed across the line so that customers all the way down to the supermarket do not get upset and buy a competitive product instead.

Understanding the importance of the problem, the CEO of the toothpaste factory gathered the top people in the company to brainstorm. They decided to start a new project, in which they would hire an external engineering company to solve their empty box problem because their own engineering department was already too stretched to take on any extra effort.

The project followed the usual process: project sponsor found, budget allocated, RFP issued, third-parties selected; 6 months (and \$180,000) later, they had a fantastic solution—on time, on budget, high quality. Everyone in the project celebrated.

They solved the problem by using high-tech precision scales that would sound a bell and flash lights whenever a toothpaste box weighed less than it should. The line would stop, and someone would yank the defective box off the line, then press another button to restart the line.

A while later, the CEO decided to review the return on investment of the project: amazing results! No empty boxes ever shipped out of the factory after the scales were put in place, and there were no related customer complaints.

“That’s some money well spent!” he said, before looking closely at the other statistics in the report.

It turns out that, after 3 weeks of production use, the number of defects picked up by the scales was zero. It should have picked up at least a dozen a day, so maybe there was something wrong with the report. He requested an inquiry, and after some investigation, the engineers reported the information was accurate. The scales were not picking up any defects because all boxes that got to that point in the conveyor belt were filled with product.

Puzzled, the CEO visited the factory floor where the precision scales were installed on the line. A few feet before the scale he saw a small desk fan blowing away the empty boxes, which were lighter than the full ones, off the belt and into a bin.

“Oh, that,” retorts one of the line workers. “We put it there because we grew tired of walking over here every time the bell rang.” Sometimes the best idea is spawned during a simple reframing.

13.3.5.2 Abstraction: Getting back to the basics

Abstraction (see [Section 2.3.3.2](#)) is the process of removing extraneous details of a work activity and focusing on the irreducible constructs of the problem to identify what is really going on, ignoring everything else. When using abstraction, look for generalizable patterns in the way particular tasks or workflows can be supported in the design. What are those commonalities, and can those patterns be defined and described so they can be reused or applied in other design situations in the ecology? What are the exceptions or variations?

13.3.5.3 Magic wand: Asking “what if?”

Another ideation technique is the magic wand technique where you ask what-if questions to temporarily set aside known constraints. This technique helps with lateral thinking and to generate ideas that, although impossible as is, shed light on other related ideas that are possible.

In the toothpaste factory example, if we had a magic wand, what would we do? We might ask for the ability to see through the cardboard box to know if it is empty. This idea of seeing through a box can now be modified to bring it out of the realm of magic by ideating on techniques to make this possible. An x-ray or sonogram machine mounted on the belt would provide the same outcome as the magical see-through idea.

13.3.5.4 Participatory design

Participatory design is a democratic process for UX design actively involving all stakeholders (e.g., employees, partners, customers, citizens, users) to help ensure that the result meets their needs and is usable. It is based on the argument that users should be involved in designs they will be using, and that all stakeholders, including and especially clients and users, have equal input into UX design (Muller & Kuhn, 1993). There have been a number of participatory design (Muller, 2003) schemes in the past to actively involve all stakeholders to get the broadest range of design ideas and to be sure all needs and perspectives are considered.

At the very beginning of a design project, you often have the user and customers on one side and system designers on the other. Participatory design is a way to combine the knowledge of work practice of the users and customers with the process, technology, and design skills of the UX and SE teams. A participatory design session usually begins with reciprocal learning, in which the users and the designers learn about each other's roles; designers learn about work practices and users learn about technical constraints (Carmel, Whitaker, & George, 1993).

In our own experience, we have found participatory design very effective for specific kinds of interaction situations. For example, we think it could be a good approach, especially if used in conjunction with design scenarios, for sketching out the first few levels of the Ticket Kiosk System interaction screens. These first screens are very important to the user experience, where first impressions are formed by users and where we can least afford to have users get lost and customers turn away. However, in our experience, the technique sometimes does not scale well to complete designs of large and complex systems. See [Section 13.7.1](#) for some history related to participatory design.

13.3.5.5 Incubation

Another kind of ideation technique is incubation. Idea incubation can occur if you take a break after a period of deep thinking about a problem. Your brain can work on the problem in the background as you back off and give it a rest, resulting in fresh perspectives after you return to the problem. Even though we are not actively working on the problem during this break, our brains have the capacity to incubate ideas in the background and come up with ideas at unexpected times, typically while doing some other activity.

13.3.5.6 *Design patterns and experience*

A design pattern is a repeatable solution to a common design problem that emerges as a best practice that encourages sharing, reuse, and consistency (see also [Sections 16.3.4](#) and [16.8](#)). Sometimes design patterns from other disciplines help trigger ideas. For example, when Rex and I were discussing the problem of the toothpaste factory, we thought of another domain that solves a different problem using weighing scales that could be used here. We remembered how Amazon packing centers use exact weight calculation to know if a package being prepared on an automated conveyor belt contains all the products the customer purchased. If the calculation is off more than a small margin of error, then the package is pushed off the conveyor belt and flagged for further analysis and human intervention. When we reuse that knowledge, we call it a design pattern, and it can be applied to the problem of the toothpaste factory, proposing a solution that pushes off a toothpaste box that does not weigh enough.

13.3.5.7 *Traverse the different dimensions of the problem space*

This ideation technique is to methodically traverse each dimension of the problem space to look for ideas that were previously missed. In this technique, designers systematically traverse the design space in search of ideas.

In the toothpaste factory example, one can traverse the different dimensions of the problem, such as mass/weight, volume, and opacity, to lead us to the following ideas:

- Mass of the product:
 - Ascertaining the mass is what it should be by weighing
 - Ascertaining the mass is what it should be through resistance to force
- Volume occupancy:
 - Internal sensor in the box
 - Displacement of fluid (air) in the box
- Seeing inside:
 - Visual inspection by manually opening each box
 - Sensing by x-ray
 - Sensing by sonogram

Note how this traversal was done by framing the problem around the toothpaste box. Another framing could be the manufacturing process

itself. Perhaps there is a different technique that wraps the box around a tube instead of building them separately and inserting one into the other?

13.3.5.8 *Seek opportunities for embodied and tangible interaction*

Embodied interaction is interaction with technology that involves a user's body in a natural and significant way, such as by using gestures (Harling & Edwards, 1997). Tangible interaction is interaction involving physical actions between human users and physical objects (Ishii & Ullmer, 1997).

Bringing physicality, tangible interaction, and embodiment into design means weaving the fabric of interaction to include not just the digital realm with its windows, icons, and menus but also the physical world where things can be touched, grabbed, turned, held, and otherwise manipulated using non-cognitive senses.

More than ever before, tangible and embodied interaction calls for physical prototypes as sketches to inspire the ideation and design process. Through physical prototypes, we can collaborate, communicate, and make meaning through physically shared objects in the real world.

In designing for embodied interaction (Tungare et al., 2006), ideate on how to involve hands, eyes, and other physical aspects of the human body in the interaction. Supplement the pure cognitive actions that designers have considered in the past and take advantage of the user's mind and body as they potentiate each other in problem solving. See [Section 13.7.2](#) for some history related to embodied and tangible interaction.

Example: Embodied and Tangible Interaction in a Board Game

If we were to make a digital version of a game, such as Scrabble, one way to do it is by creating a desktop application, where people operate in their own window to type in letters or words. This makes it an interactive game but not embodied.

Another way to make Scrabble digital is the way Hasbro did it in Scrabble Flash ([Fig. 13.4](#)) (Hasbro Games, 2011). Hasbro Games used embedded technology in producing an electronic version of the old board game. They made the game pieces into real physical objects with built-in technology. Because players hold the SmartLink letter tiles in their hands, it makes them fundamentally embodied and tangible.

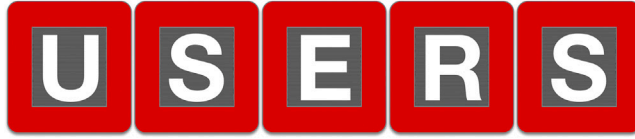


Fig. 13.4

Approximate appearance of Scrabble Flash, a game by Hasbro, Inc.

At the start of a player's turn, the tiles each generate their own letter for the turn. The tiles can read each other's letters if they touch as a player physically shuffles them around. When the string of between two and five letters makes up a word, the tiles light up and beep, and the player can try for another word with the same tiles until time is up. The tiles also work together to time each player's turn, flag duplicates, and display scores. It also has a built-in dictionary as an authority (however arbitrary it may be) on what comprises a real word.

13.3.6 Doing Ideation

Ideation as a group activity. Even though ideation can be performed by designers on their own, it can be more effective when undertaken as a group with members from different backgrounds operating in an immersive environment, such as your UX studio. When undertaken as a group activity, it is a kind of brainstorming, often starting with an overview discussion to establish background and parameters and agreement on goals of the ideation exercise.

Mechanics of ideation. The goal here is to have intense rapid interactions to spawn and accumulate large numbers of design ideas about characteristics and features. Write ideas on a whiteboard or use marking pens on flip charts. Or write your ideas on sheets of paper so that you can move the sheets around to organize them.

Make use of UX research models. The social, flow, artifact, and physical models are all visual, which makes them ideal for informing and inspiring design. We suggest that you organize some of your ideation sessions around each of these models by making each the focal point of design thinking and ideation before moving on to the next.

Make use of ideation techniques (Section 13.3.5). For example, ask yourself what design you would create if you had a magic wand. Look at the abstract work activity (see Section 12.3.6) to see what ideas it triggers.

Use teamwork and play off each other's ideas while living the part of the user.

Talk in scenarios, keeping customers and users in the middle, telling stories of their experience as your team weaves a fabric of new ideas for design solutions.

Make the outputs of your ideation as visual and tangible as possible. Intersperse an outline with multiple rounds of sketches. Post and display everything all around the room as your visual working context. Where appropriate, build physical mockups as embodied sketches. You can include examples of other systems, conceptual ideas, considerations, design features, marketing ideas, and experience goals. Get all your whacky, creative, and off-the-wall ideas out there. The flow should be a mix of verbal and visual. When the fountain of new ideas seems to have run dry for the moment, the group can switch to critiquing mode.

Example: Ideation at IDEO

In IDEO's "deep dive" approach (Brown, 2008), a cross-disciplinary group works in total immersion without consideration of rank or job title. In their modus operandi of focused chaos, "enlightened trial and error succeeds over the planning of lone genius." Their design process was illustrated in a well-known ABC News documentary (ABC News Nightline, 1999) with a new design for supermarket shopping carts, beginning with a brief UX research inquiry where team members visited different stores to understand the work domain of shopping and issues with existing shopping cart designs and use.

Then, in an abbreviated UX research synthesis process, they regrouped and engaged in debriefing, synthesizing different themes that emerged in their UX research inquiry. This analysis fed parallel brainstorming sessions in which they captured all ideas, however unconventional. At the end of this stage, they indulged in another debriefing session, combining the best ideas from brainstorming to assemble a physical design prototype. This alternation of brainstorming, prototyping, and review, driven by their "failing often to succeed sooner" philosophy, is a good approach for anyone wishing to create a good user experience.

Example: Ideation for the Ticket Kiosk System

We brainstormed with potential ticket buyers, students, Middleburg University (MU) representatives, and civic leaders. Here, we show a greatly abbreviated list of topics covered over multiple sessions at various times. A more complete list is available on the book website (<https://www.theuxbook.net/>).

Thought questions to get started:

What does “an event” mean? How do people treat events in real life?

An event is more than something that happens and maybe you attend.

An event can have emotional meanings, can be thought provoking, can have meaning that causes you to respond and act.

Data artifacts:

Tickets, events, event sponsors, MU student ID, kiosk

Things people may want to do with tickets:

Email tickets to friends

Possible features and breadth of coverage:

Customized tickets for keepsake editions

Homecoming events

Parents’ weekend events

Visiting speakers on current topics

Visitor’s guide to what’s happening in town and at MU

Christmas tour of Middleburg

View Christmas decorations on historic homes

Walk Main Street to see decorations and festive shops

Exercise 13.1 Ideation About Aircraft Flight Recorders

This is a team generative design exercise, so gather into your team, go to a quiet place (your UX studio, if you have one), and begin ideation and sketching, using what you know of the domain.

When planes crash, we often hear that the flight recorder box, which has the most recent flight and pilot data on magnetic tape, is the key to understanding the cause. However, we also often hear that the flight recorder box cannot be found or is found damaged in the crash.

Considering available technology and the broadest context and ecosystem in which planes and flight recorders operate, develop a better conceptual design than what now exists.

13.4 SKETCHING

Going hand in hand with ideation, sketching as an indispensable part of design goes back at least to the Middle Ages. Consider da Vinci and his famous sketchbooks. As Buxton (2007b) puts it, if you are not sketching, you are not doing design.

13.4.1 Characteristics of Sketching

Sketching is the rapid creation of freehand drawings expressing preliminary design ideas, focusing on concepts rather than details. We credit Buxton (2007b) as the champion for sketching, who has been the impetus for most of this topic here.

Following are more defining characteristics of sketching (Buxton, 2007b; Tohidi et al., 2006):

- Everyone can sketch; you do not have to be artistic.
- Most ideas are conveyed more effectively with a sketch than with words.
- Sketches are quick and inexpensive to create; they do not inhibit early exploration.
- Sketches are disposable; there is no real investment in the sketch itself.
- Sketches are timely; they can be made just in time, done in the moment, provided when needed.
- Sketches should be plentiful; entertain a large number of ideas and make multiple sketches of each idea.
- Textual annotations play an essential support role, explaining what is going on in each part of the sketch and how.

13.4.1.1 Sketching is essential to ideation and design

Sketching is an indispensable part of design. Nilsson and Ottersten (1998) describe sketching as an essential visual language for brainstorming and discussion. Design is a process of creation and exploration, and sketching is a visual medium for exploration.

Sketching captures ideas in an embodied and tangible way; it externalizes the mental description of an idea for sharing, analysis, and archiving. By opening up new pathways to create new ideas, sketching acts as a multiplier in ideation. Sketching adds visualization to ideation, supercharging cognition and boosting creativity by bringing more human senses to the task (Buxton, 2007b).

13.4.1.2 Sketching is a conversation about user experience

Sketching is not art. Sketching is not about putting pen to paper in the act of drawing, nor is it about artistic ability. A sketch is not about making a true-to-life drawing or picture of a product.

A sketch is a conversation. A sketch is not an artifact that you look at, but a conversation about design. Your sketches represent your thoughts about a

design. A sketch is also a vehicle for communicating design ideas among the design team.

Sketches can record design history. A series of sketches can document the evolution of a design over time.

A sketch is about the user experience, not the product. In a talk at Stanford, Buxton (2007a) challenged the audience to draw his mobile phone, but not a drawing of the phone as a product. He meant something much more difficult: a sketch that reveals the interaction or the experience of using the phone in a situated context, where the product and its physical affordances encourage one type of behavior and experience over another.

A sketch is not expected to look like a finished product. Sketches should be intentionally rough and open to interpretation.

13.4.1.3 Sketching is embodied cognition to aid invention

Designers invent while sketching. A sketch is not just a way to represent your thinking; the act of making the sketch is part of the thinking (Fig. 13.5). In fact, the sketch itself is less important than the process of making it.

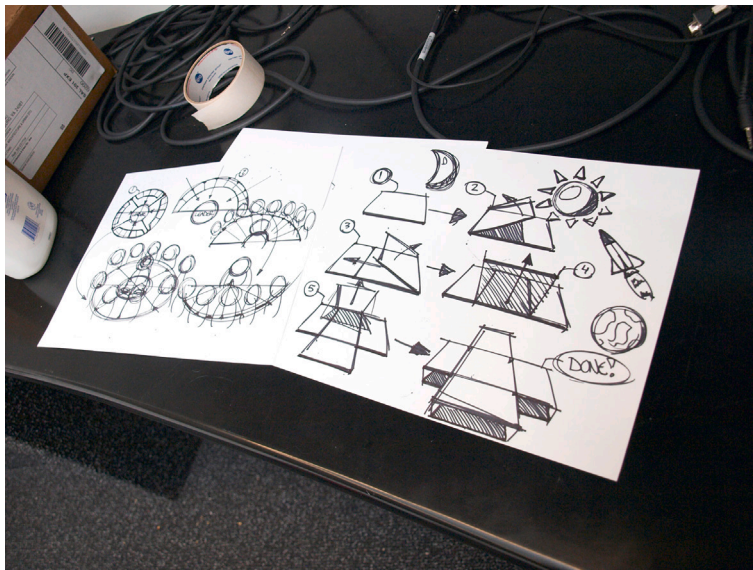


Fig. 13.5

Sketches to think about design. (From Akshay Sharma, Virginia Tech Department of Industrial Design.)

The importance of involving your hands in sketching. The kinesthetics of sketching, pointing, holding, and touching bring the entire hand-eye-brain coordination feedback loop to bear on the problem solving. Your physical motor movements are coupled with perceptual and cognitive activity; the designer’s mind and body potentiate each other in invention (Baskinger, 2008).

13.4.2 Doing Sketching

13.4.2.1 Stock up on sketching and mockup supplies

Stock the ideation studio with sketching supplies, such as sketchbooks, whiteboards, blackboards, corkboards, flip chart easels, Post-it labels, tape, and marking pens. Be sure to include supplies for constructing physical mockups, including scissors, hobby knives, cardboard, foam core board, duct tape, wooden blocks, push pins, string, bits of cloth, rubber, other flexible materials, crayons, and spray paint.

13.4.2.2 Use the language of sketching

The vocabulary of sketching. To be effective at sketching for design, you will use a vocabulary that has remained mostly the same for centuries. One important language feature is the vocabulary of lines, where lines are made as freehand open gestures. Instead of being mechanically correct and perfectly straight, lines in sketches are roughed in and not connected precisely. In this language, lines overlap and often extend beyond the corner. Sometimes they miss intersecting and leave the corner slightly open.

An unfinished appearance invites exploration. The low resolution and ongoing nature of a sketch suggest it is a concept in the making, not a finished design. It needs to look disposable and inexpensive to make. Sketches are deliberately ambiguous and abstract, leaving space for the imagination about other aspects of the design. You can see this unfinished look in the sketches of [Figs. 13.6 and 13.7](#).

Keep sketches open to interpretation. Effective sketches are impressionistic in that they are open to interpretation, fostering new relationships to be seen within them, even by the person who drew them. To support ideation, a sketch should propose exploration, saying “Let’s play with this and see what comes up.”

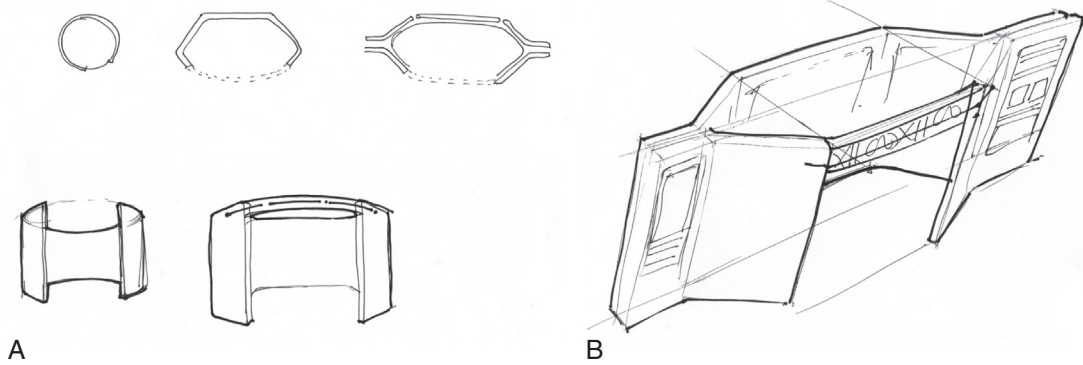


Fig. 13.6
Freehand gestural sketches
for the Ticket Kiosk System.
(From Akshay Sharma,
Virginia Tech Department of
Industrial Design.)

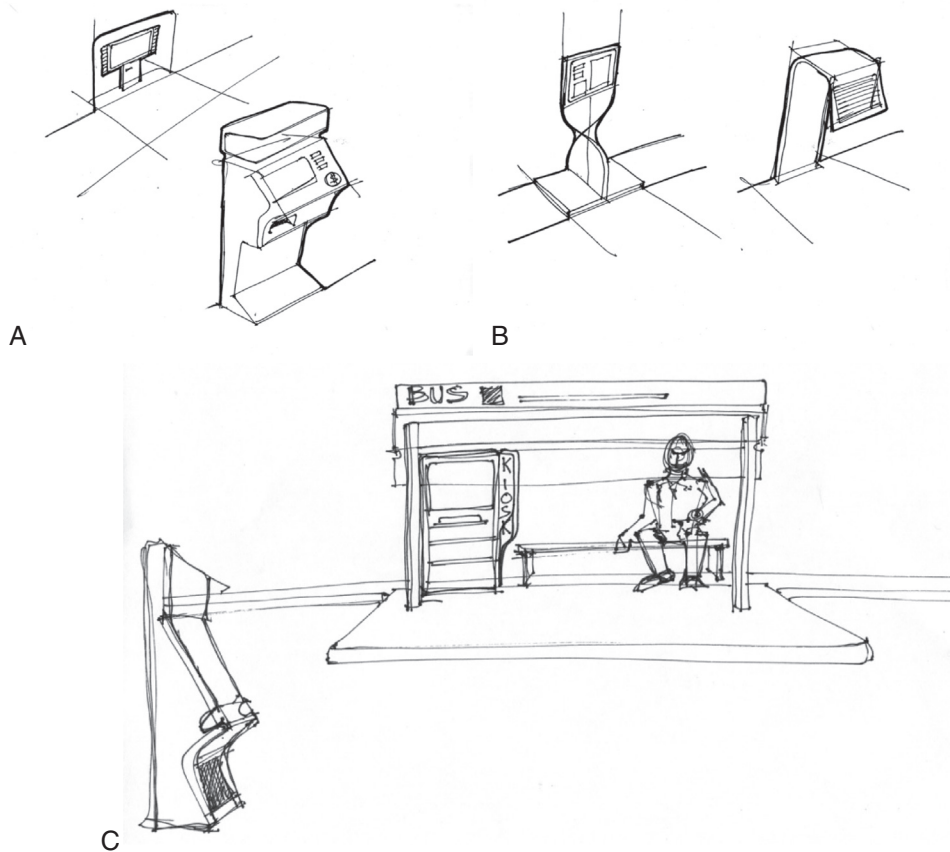


Fig. 13.7
Ideation and design
exploration sketches
for the Ticket Kiosk
System. (From Akshay
Sharma, Virginia
Tech Department of
Industrial Design.)



Fig. 13.8

*Designers doing sketching.
(From Akshay Sharma,
Virginia Tech Department of
Industrial Design.)*

Fig. 13.8 provides examples of what the sketching activity looks like in a real design studio.

Exercise 13.2 Practice in Ideation and Sketching

Goal: To practice ideation and sketching for design

Activities: Work in a small group is strongly preferable, but you also can work with one other person.

- Gather blank paper, appropriate marking pens, and any other supplies you may need for sketching.
- Pick a topic, system, or device. Our recommendation is something familiar, such as a dishwasher.
- Begin with free-flow ideation about ways to design a new and improved concept of a dishwasher. Do not limit yourself to conventional designs.
- Go with the flow and see what happens.
- Remember that this is an exercise about the process, so what you come up with for the product is not crucial.
- Everyone should make sketches of the ideas that arise about a dishwasher design as you go in the ideation.
- Begin with design sketches in the ecological perspective. For a dishwasher, this may include your dining room, kitchen, and the flow of dishes in their daily cycle. You could include something unorthodox (e.g., a conveyor belt from the dinner table through your appliance and out into the dish cabinets). Sketch how avoiding the use of paper plates can save resources and not fill the trash dumps.

- Make some sketches from an interaction perspective showing different ways you can operate the dishwasher (e.g., how you load and unload it, how you set wash cycle parameters and turn them on).
- Make sketches that project the emotional perspective of a user experience with your product. This may be more difficult, but it is worth taking some time to try.
- Ideate. Sketch, sketch, and sketch. Brainstorm and discuss.

Deliverables: A brief written description of the ideation process and its results, along with all your supporting sketches.

Schedule: Give yourself enough time to be fully engaged in this activity.

13.4.3 Physical Mockups as Embodied Sketches

Just as sketches are two-dimensional vehicles for invention, a physical mockup for ideation about a physical device or product is a three-dimensional sketch. Physical mockups as sketches, like all sketches, are made quickly, are highly disposable, and are made from at-hand materials to create tangible props for exploring design visions and alternatives.

A physical mockup is an embodied sketch because it is even more of a physical manifestation of a design idea, and it is a tangible artifact for touching, holding, and acting out usage (Fig. 13.9).

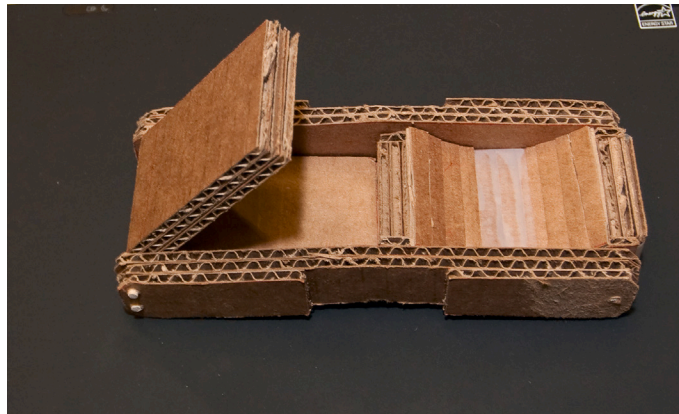


Fig. 13.9

Example of rough physical mockup of a rickshaw vehicle for personal transportation. (From Akshay Sharma, Virginia Tech Department of Industrial Design.)

For later in the process, after design exploration is complete, you may want a more finished-looking three-dimensional design representation (Fig. 13.10) to show clients, customers, and implementers.



Fig. 13.10

Example of a more finished-looking physical mockup. (From Akshay Sharma, Virginia Tech Department of Industrial Design.)

13.5 CRITIQUING

Critiquing is the activity where design ideas are assessed to identify advantages, disadvantages, and constraints and tradeoffs.

At a high level, the goal of critiquing is to analyze if the design:

- Meets design goals
- Fits well with the ecology (i.e., communicates seamlessly with other devices and systems in the environment)
- Supports the interaction needed with other devices
- Provides good usability
- Evokes positive emotional impact
- Engenders meaningfulness for users

This evaluation part of generative design is never formal; there are no established methods. It is a fast, furious, and freewheeling comparison of many alternatives and inspiration for further alternatives.

13.5.1 Include Users in the Critiquing Activity

Pull together the best of all the different brainstorming ideas into a single big-picture redesign of how work gets done and communicate the new

vision to customers and users. For this critical analysis, bring in a cross section of people with broadly varying backgrounds, perspectives, and personalities.

- Show off your ideas and sketches along with other mockups and models.
- Let them talk, argue, and criticize.
- Your job is to listen.

Fig. 13.11 provides an example of critiquing in midstream within the Virginia Tech Kiva, an ideation studio.



Fig. 13.11
Critiquing within the
Virginia Tech ideation
studio, the Kiva. (From
Akshay Sharma, Virginia
Tech Department of
Industrial Design.)

13.6 RULES OF ENGAGEMENT FOR IDEATION, SKETCHING, AND CRITIQUING

13.6.1 Behave Yourself

The process should be democratic, which indicates the following:

- Every idea has equal value.
- The process is ego free.
- No individual takes ownership of ideas.
- All ideas belong to the group.

13.6.2 Be Aware of Which Mode You Are In

Back in graduate school, I (Rex) read a simple paper on “stop-and-go thinking” (Mason, 1968) that has affected my outlook on problem solving and brainstorming ever since. This paper suggests that, if you keep the creative mindset of ideation separate from the judicial mindset of critiquing, you will get better ideas and make wiser judgments.

Although you will interweave ideation and critiquing throughout the design process, you should know which mode you are in at any given time and not mix the modes. Ideation should result in a pure flow of ideas regardless of feasibility, in the classic tradition of brainstorming. Although we know that, at the end of the day, practical implementation constraints must be considered and allowed to carry weight in the final overall design, saying “Hey, wait a minute!” too early can stifle innovation.

Mason (1968) calls this separation of ideation and critiquing “go-mode and stop-mode thinking.” In idea-creation (go) mode, you adopt a freewheeling mental attitude that will permit ideas to flourish. In critiquing (stop) mode, you revert to a cold-blooded, critical attitude that will bring your judgment into full play.

Ideation gives you permission to be radical; you get to play outside the safe zone yet remain safe. Allowing early cries of “that will never work,” “they have already tried that,” “it will cost too much,” “we do not have a widget for that,” or “it will not work on our implementation platform” will unfairly hobble and frustrate this first step of creativity.

The design teams at IDEO (ABC News Nightline, 1999) signal premature critiquing by ringing a wrist-mounted bicycle bell to signal the foul of being judgmental too early in design discussions.

13.6.3 Iterate to Explore

Finding the best design ideas requires exploring a large number of possibilities and candidate designs. The best tools for exploring ideas are ideation, sketching, and critiquing, combined with extensive iteration (Buxton, 2007b). Be ready to try, try, try, and try again, rapidly. Think about Thomas Edison and his more than 10,000 experiments to create a usable and useful lightbulb.

Iteration is that magical process through which you undertake course changes to understand the real design problem and solutions. Frequent iteration is the key to getting feedback to improve your designs in agile processes. See [Chapter 4](#) for much more about the necessity of iteration in most aspects of UX design.

13.7 RELATED HISTORY

13.7.1 History Related to Participatory Design

The idea of user participation in system design harkens back (as does UX research) at least to a body of effort called work activity theory (Bødker, 1991; Ehn, 1990). Originating in Russia and Germany, it flourished in Scandinavia in the 1980s, where it was closely related to the workplace democracy movement. These early versions of participatory design embraced a view of design based on work practice situated in a worker's own complete environment, but also espoused empowerment of workers to “co-determine the development of the information system and of their workplace” (Clement & Besselaar, 1993).

Going back to the 1980s and earlier, probably the most well-known participatory design project was the Scandinavian project called UTOPIA (Bødker et al., 1987). A main goal of Project UTOPIA was to overcome limitations on opportunities for workers to affect workplace technology and organizational work practices. UTOPIA was one of the first such projects intended to produce a consumer product at the end of the day.

In the early 1990s, the Scandinavian approach to democratic design was adapted and extended within the human–computer interaction (HCI) community in the form of participatory design. Muller's vision of participatory design as embodied in his PICTIVE³ approach (Muller, 1991; Muller, Wildman, & White, 1993b) is the most well-known adaptation of the general concept specifically to HCI. The first Participatory Design Conference met in 1990, and it has been held biannually ever since. Participatory design has since been codified for practice (Greenbaum & Kyng, 1991), reviewed (Clement & Besselaar, 1993), and summarized (Muller, 2003).

13.7.2 History Related to Embodied and Tangible Interaction

Tangible interaction is a key area of focus in industrial design, pertaining to designing objects and products to be held, felt, and manipulated by humans. Embodied and tangible interactions are closely related. Clearly, this is also closely related to the idea of physicality (see [Section 27.3.7](#)), which pertains to direct physical interaction by a person with real physical (hardware) devices as in the grasping and moving of knobs and levers.

Simply put, embodiment means “having a body.” So, taken literally, embodied interaction is about using one's physical body while interacting

³Plastic Interface for Collaborative Technology Initiatives through Video Exploration

with the surrounding technology. As Dourish (2001) explains, however, embodiment does not simply refer to physical reality but “the way that physical and social phenomena unfold in real time and real space as a part of the world in which we are situated, right alongside and around us.”

As a result, embodiment is not about people or systems per se. Dourish (2001) stated that “embodiment is not a property of systems, technologies, or artifacts; it is a property of interaction. Cartesian approaches separate mind, body, and thought from action, but embodied interaction emphasizes their duality.”

Although tangible interaction (Ishii & Ullmer, 1997) seems to have a following of its own, it is very closely related to embodied interaction. You could say that they are complements to each other. Tangible design is about interactions between human users and physical objects. Industrial designers have been dealing with it for years, designing objects and products to be held, felt, and manipulated by humans. The difference now is that the object involves some kind of computation. Also, there is a strong emphasis on physicality, form, and tactile interaction (Baskinger & Gross, 2010).

13.8 SEGUE

In this chapter, we discussed the generative design activities of ideation, sketching, and critiquing. In the next chapter, we look at ways to communicate these design ideas through conceptual designs that aid in creating effective mental models for users.